Master Serology and Vaccine Carrier Table

Compiled from Douglas's Google Drive on November 5, 2026. Sources: Unified Vaccine and Carrier Architecture Master Reference (V1 and V2), Lancefield Antigen Groups, the fungal analogues note, and the Meta/Gemini test scripts.

How to read this table

Serotype, serogroup, lineage, vaccine coverage and disease syndrome are kept separate. A shared carrier protein means a shared manufacturing tool, not shared biology or cross-protection. Evidence status is my own assessment: Established means licensed and well documented, Investigational means in trials, Experimental means preclinical or early research, Verify means the source is AI-generated or secondary and needs a primary-literature check.

Table 1. Bacterial capsular and toxin systems

Disease	Pathogen	Type	Serotyping or antigen system	Antigen chemistry	Genetic locus	Vaccines	Carrier	Evidence status
Diphtheria	Corynebacterium diphtheriae	Gram-positive rod, toxigenic	No capsular serotypes; biotypes gravis, mitis, intermedius; tox+ vs tox-	Diphtheria toxin protein	tox gene on corynephage beta; dtxR regulator	DT, Td, DTaP, Tdap, Pediarix, Pentacel, Infanrix hexa	Toxoid is the antigen; source of CRM197	Established
Tetanus	Clostridium tetani	Anaerobic spore former	Toxin only, no serotypes	Tetanospasmin protein	tent gene on plasmid	TT, DT, Td, DTaP, Tdap	Toxoid is the antigen; also used as carrier TT	Established
Pneumococcal disease	Streptococcus pneumoniae	Encapsulated Gram-positive coccus	100+ capsular serotypes by Quellung reaction	Unique repeat unit per serotype; 19F phosphodiester bridge; 6A and 6B ribitol-phosphate	cps locus between dexB and aliA; wzx, wzy; wciP SNP at 584 separates 6A from 6B	PCV7, PCV10 (Synflorix, Pneumosil), PCV13, PCV15, PCV20, PPSV23	CRM197 for Prevnar family and Pneumosil; Protein D, TT (18C), DT (19F) in Synflorix; PPSV23 has none	Established
Meningococcal disease	Neisseria meningitidis	Encapsulated Gram-negative diplococcus	Serogroups A, B, C, W, Y, X by capsule; serotypes by PorB	A: ManNAc-1-phosphate; C: sialic acid alpha 2-9; B: sialic acid alpha 2-8; W and Y: sialic acid with galactose or glucose	cps locus; sia genes	Menactra, Menveo, MenQuadfi, Nimenrix, Menjugate, NeisVac-C, MenAfriVac, Bexsero, Trumenba	DT, CRM197, TT; MenB uses protein or outer membrane vesicle antigens	Established

Disease	Pathogen	Type	Serotyping or antigen system	Antigen chemistry	Genetic locus	Vaccines	Carrier	Evidence status
Haemophilus influenzae type b	Haemophilus influenzae	Encapsulated Gram-negative coccobacillus	Serotypes a to f; type b most invasive; NTHi has no capsule	PRP (polyribosylribitol phosphate)	cap locus; bexABCD export	ActHIB, Hiberix, HbOC, PedvaxHIB, Menitorix	TT, CRM197, OMP	Established
Typhoid fever	Salmonella Typhi	Gram-negative rod	Vi positive, O9, H:d	Vi polysaccharide, poly alpha 1-4 GalNAcA	viaB locus in SPI-7	Typhim Vi (no carrier), Typbar-TCV (Vi-TT); Vi-CRM investigational	TT, CRM197	Established for Vi-TT; Vi-CRM investigational
Group B streptococcal disease	Streptococcus agalactiae	Encapsulated Gram-positive coccus	Serotypes Ia, Ib, II, III, IV, V, up to IX	Sialic acid-containing capsular polysaccharides	cps locus (cpsA to cpsK), organized like pneumococcus	None licensed; GBS6 and GBS-NN investigational	CRM197, TT, native proteins	Investigational
Non-typhoidal salmonellosis	Salmonella Enteritidis and Typhimurium	Gram-negative rod	Kaufmann-White scheme, O and H antigens	LPS O-antigen repeats	rfb cluster; fliC, fljB	None for humans; veterinary vaccines exist	TT, CRM197, DT in experimental conjugates	Investigational
Paratyphoid fever	Salmonella Paratyphi A	Gram-negative rod	O2, O12, H:a	O-specific polysaccharide	wba and rfb clusters	O-SP conjugates investigational	TT, CRM197	Investigational

Table 2. Fungal and viral comparison entries

Disease	Pathogen	Antigen structure	Genetic locus	Vaccine status	Role in the framework	Evidence status
Cryptococcosis	Cryptococcus neoformans and gattii	Glucuronoxylomannan capsule with mannose backbone, xylose and glucuronic acid side chains; serotypes A to D	CAP10, CAP59, CAP60, CAP64	No licensed vaccine; experimental GXM conjugates with TT, DT, CRM197	Fungal capsule that is analogous in function to bacterial capsules	Experimental
Candidiasis	Candida albicans and auris	Cell wall mannan, beta-glucan, phosphopeptidomannan, Als3 adhesin	ALS gene family	NDV-3A and PEV7 investigational	Surface polymers that trigger pattern recognition receptors	Experimental; verify
Aspergillosis	Aspergillus fumigatus	Galactomannan cell wall component and diagnostic biomarker	Not stated in sources	None	High-molecular-weight polysaccharide that resists clearance	Experimental; verify

Disease	Pathogen	Antigen structure	Genetic locus	Vaccine status	Role in the framework	Evidence status
Influenza	Influenza virus A and B	Hemagglutinin and neuraminidase glycoproteins	8 RNA segments	Inactivated, live attenuated, recombinant	Contrast: viral glycoprotein, not a polysaccharide serotype	Established
HPV disease	Human papillomavirus	L1 capsid virus-like particles	L1; E6 and E7 oncogenes	Gardasil 4, Gardasil 9, Cervarix	Contrast: no carrier needed	Established

Table 3. Lancefield streptococcal groups

Group	Main organisms	Typical significance	Evidence status
A, B, D	Documented in the existing reference manual	Core clinical groups	Established
C and G	Streptococcus dysgalactiae subsp. equisimilis, Streptococcus equi	Pharyngitis, skin and soft tissue infection, bacteremia, toxic shock syndrome	Verify against primary sources
F	Streptococcus anginosus group	Deep tissue abscesses, postsurgical infection	Verify
H and K	Streptococcus sanguinis (H), Streptococcus salivarius (K)	Oral commensals	Verify
N	Lactococcus lactis	Dairy fermentation	Verify
R and S	Streptococcus suis	Swine pathogen with zoonotic spread	Verify
E, L, M, O	Named in the source but not detailed	Animal and other species	Gap: no detail in the files

Diagnostic latex kits routinely screen groups A, B, C, D, F and G together.

Table 4. Carrier protein map

Carrier	Pneumococcal	Meningococcal	Hib	Typhoid	Other
CRM197	Prevnar 13, Prevnar 20, Vaxneuvance, Pneumosil	Menveo, Menjugate	HbOC	Vi-CRM (investigational)	GBS6, Salmonella and Cryptococcus conjugates (investigational)
Diphtheria toxoid (DT)	Synflorix 19F	Menactra	Historical	Investigational	Experimental conjugates
Tetanus toxoid (TT)	Synflorix 18C	MenQuadfi, Nimenrix, MenAfriVac, Menitorix	ActHIB, Hiberix, Menitorix	Typbar-TCV	Experimental conjugates
Protein D (from NTHi)	Synflorix, 8 serotypes	None listed	None listed	None listed	None
OMP or OMV	None	Bexsero (MenB)	PedvaxHIB	None	None

Table 5. Combination vaccine platforms

Platform	Diseases covered	Architecture
DT, Td, DTaP, Tdap	Diphtheria, tetanus, pertussis; with polio, Hib, hepatitis B in Pediarix, Pentacel, Infanrix hexa	Toxoids serve as antigen and carrier backbone
Menitorix	Meningococcal C and Hib	Both polysaccharides conjugated to tetanus toxoid
MMR and MMRV	Measles, mumps, rubella, varicella	Live attenuated viruses
Twinrix	Hepatitis A and B	Inactivated plus recombinant subunit

Table 6. The wider archive, by project

Project	Contents	Status
Unified Relativity Architecture (Book 11)	Schoonover Field extension, 361-degree phase operator, energy-consequence equation Ec = sum of wi mi vi to the x	Active authoring
Compendium Books 1 to 4	A Pathway to the Cure; Quantum Thinking; Chronomedicine and Nameology; Clinical Diagnostic Technician Manual	Short drafts, each with a parallel backup and verification manifest
Doug's Thoughts on Teaching Medicine	Non-fiction textbook draft	Draft
Pathway to the Cure: Quantum Medicine	Theoretical reference manuscript	Draft
The Middle: Ether, Osmosis, and the Contraction Event	Theoretical framework paper	Completed
Serology research track	Master Reference V1 and V2, Lancefield groups, fungal analogues	Active
Ember and Muse workspace	SOUL.md, personality file, shared research bundle, Meta and Gemini test scripts	Active
Grand Export	Index of 373 works in a single document of about 288,000 characters	Stage 1 complete

Open gaps and cautions

- Three near-duplicate copies of the Master Reference exist in Drive.
- The Lancefield document ends with raw import code that should be removed.

- Five Muse links in the research bundle expire November 6 at about 7:42 PM Central, and I could not open them.
- Fungal and Lancefield entries come from AI-generated or secondary sources and need primary verification.
- Wording such as "links the diphtheria name to pneumococcal 19F" or "proves fungal capsules use identical chemistry" overstates what shared carriers show; shared carriers do not establish cross-protection.
- Disease rankings should always name a population and time period.